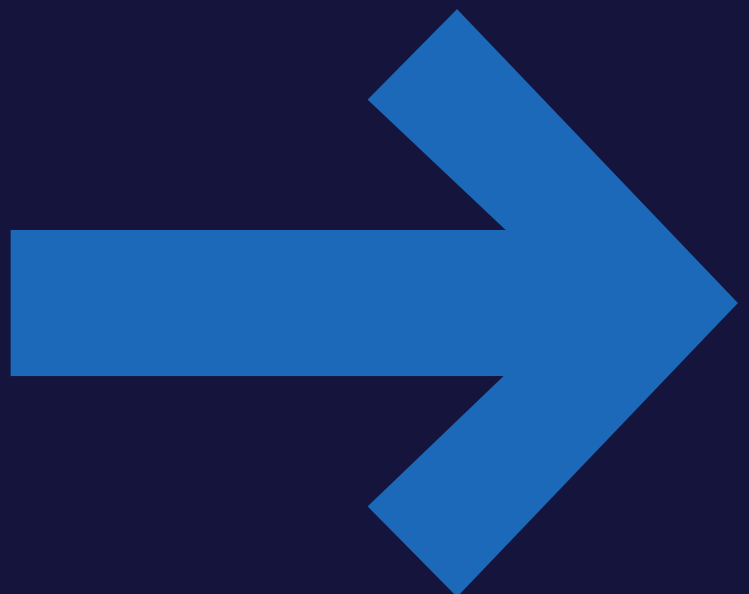


# Notification Broker

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## AS4-Gateway

Version: 1.1 - date: 02-03-2026



# Notification Broker

## Document change history

| Version | Description                                                                                                                                                           | Who            | When       |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|
| 1.0     | First version                                                                                                                                                         | Ulrik Andersen | 06-02-2026 |
| 1.1     | Added section describing different scenarios regarding downtime.<br>Corrected description of the 'Role' child elements.<br>Changed 'operator' to 'Economic Operator'. | Ulrik Andersen | 02-03-2026 |
|         |                                                                                                                                                                       |                |            |

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## Background

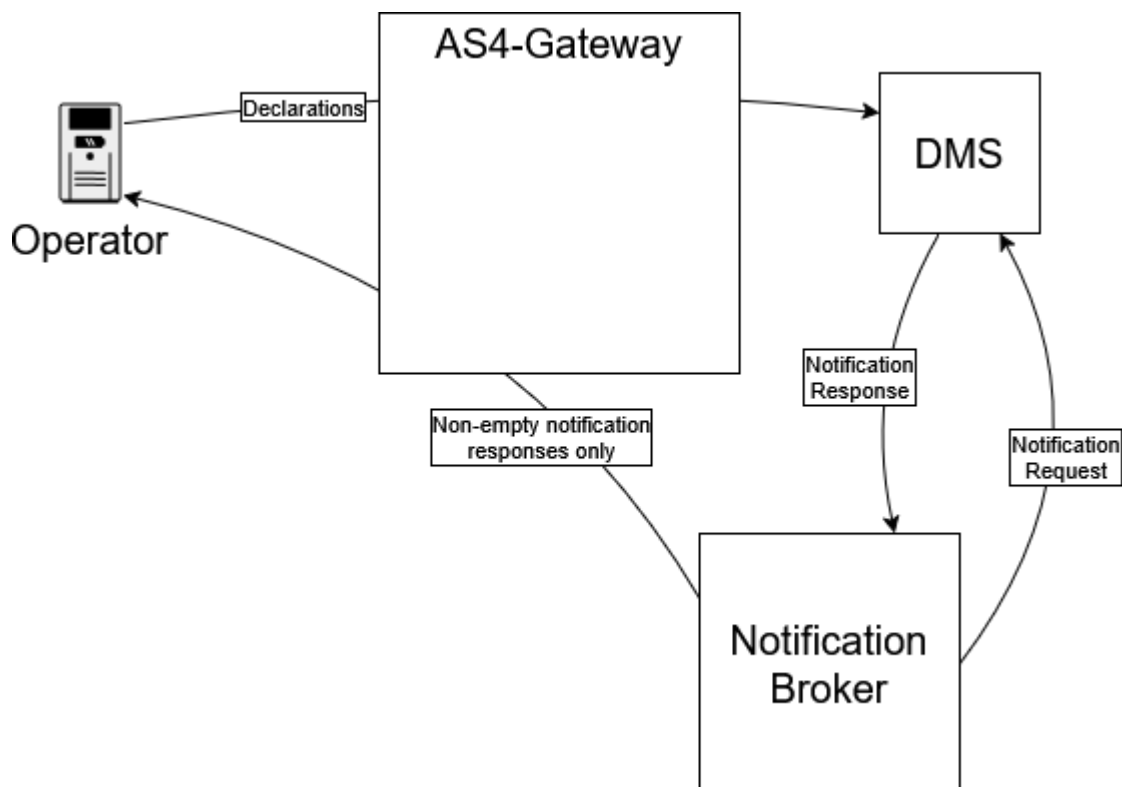
When using DMS System-to-System, notifications for a specific period are fetched from DMS by sending a notification request to the AS4-Gateway. As notifications are generated by the DMS system anytime during the day, then each Economic Operator must ensure that the periods in the notifications requests cover the whole day. E.g. using an interval of 2 minutes, the periods (fromDate – toDate) would be as follows:

- ...
- 12:00:00 – 12:02:00
- 12:02:00 – 12:04:00
- 12:04:00 – 12:06:00
- ...

Currently the notification requests amounts to around 70% of all the requests towards the AS4-Gateway, and only between 3% - 10% of the responses contain notifications. This means that most of the traffic between the Economic Operators and the AS4-Gateway is essential “waste of bandwidth and client and server resources”.

The Notification Broker has been developed to address this issue. The Notification Broker will send notification requests to DMS on behalf of the Economic Operator, and the response will only be sent to the Economic Operator if it contains notifications.

Overview of message flow:



This solution gives the following benefits:

- Significantly reduces the traffic between the Economic Operators and the AS4-Gateway.
- It will ease the setup of system-to-system integration with DMS, as it will no longer be required to send notification requests.

- Combining the Notification Broker with “push of responses”, then Economic Operators only need to send declarations, and all responses (notifications, errors, etc.) will be pushed from the Told AS4-Gateway to the Economic Operator.

## Solution in detail

### Configuration

To configure an Economic Operator to receive notifications via the Notification Broker, the following information is required:

#### AS4-Gateway username

The username used for system-to-system integration with DMS. The username consists of a CVR-number and a certificate UID. Example value:

- CVR\_34730466\_UI\_e7057bbd-5e7e-4300-ad81-623333d5c657

#### Start time, interval and overlap

The Notification Broker uses 3 values to calculate the period in the notification requests (fromDate/toDate):

- Start time
- The interval in seconds
- Overlap in seconds

Example: If the start time is “01-02-2026 12:00:00” and the interval and overlap is 120 seconds and 10 seconds respectively, then the Notification Broker would send the following notification requests.

| fromDate            | toDate              |
|---------------------|---------------------|
| 2026-02-01T11:59:50 | 2026-02-01T12:02:00 |
| 2026-02-01T12:01:50 | 2026-02-01T12:04:00 |
| 2026-02-01T12:03:50 | 2026-02-01T12:06:00 |
| ...                 | ...                 |

The notification request will be sent after the calculated toDate.

### Services

The services to request notifications for. This can be one or more of the following services:

- DMS.Import2.Notification
- DMS.Export2.Notification
- DMS.Transit.Notification

### Notification responses

The notification responses will be sent to the Economic Operator in the same way as responses to notification requests generated and sent by an Economic Operator. There are however some differences in the returned AS4 User Message:

- The property RefToOriginalMessageId will not be present as there is no original notification request sent by the Economic Operator.
- Some additional properties are added, so the Economic Operator can identify the period the notifications belong to and whether the response is part of a multi-page response.  
See below for details.

The table below gives an overview of the returned child elements in the UserMessage element and whether the element/value differs from a response to a notification request sent by an Economic Operator.

| Element | Description | Difference |
|---------|-------------|------------|
|---------|-------------|------------|

|                                  |                                                                                                                                    |                                                                                                                                                       |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Timestamp                        | Time when the AS4 message was generated                                                                                            | No                                                                                                                                                    |
| MessageId                        | Unique ID identifying the AS4 message                                                                                              | No                                                                                                                                                    |
| From                             | The sender of the message.                                                                                                         | The value of the 'Role' child element is set to Initiator (see example in Appendix)                                                                   |
| To                               | The receiver of the message.                                                                                                       | The value of the 'Role' child element is set to Responder (see example in Appendix)                                                                   |
| Service                          | "http://docs.oasis-open.org/ebxml-msg/as4/200902/service"                                                                          | No                                                                                                                                                    |
| Action                           | "Response"                                                                                                                         | No                                                                                                                                                    |
| ConversationId                   | Contains a unique ID identifying the notification response. All pages in a multi-page response will have the same conversation ID. | Yes, for Economic Operator-triggered responses this value is taken from the ConversationId in the notification request sent by the Economic Operator. |
| Property: fromDate               | The fromDate of the notification request sent by the Notification Broker.                                                          | Not present for notification responses triggered by an Economic Operator.                                                                             |
| Property: toDate                 | The toDate of the notification request sent by the Notification Broker.                                                            | Not present for notification responses triggered by an Economic Operator.                                                                             |
| Property: page                   | The requested page in the notification request sent by the Notification Broker.                                                    | Not present for notification responses triggered by an Economic Operator.                                                                             |
| Property: size                   | The requested size in the notification request sent by the Notification Broker.                                                    | Not present for notification responses triggered by an Economic Operator.                                                                             |
| Property: sequenceNumber         | A sequence number, starting with 1, per system (Transit, Export, Import).                                                          | Not present for notification responses triggered by an Economic Operator.                                                                             |
| Property: RefToOriginalMessageId | N/A                                                                                                                                | Only present for notification responses triggered by an Economic Operator.                                                                            |

## Switch to Notification Broker

### Switch-over plan

An Economic Operator, who wants to switch over to the Notification Broker instead of sending notification requests from their own system, must do the following:

1. Send a request to Told Service Desk with the information specified in the section "Configuration":
  - a. AS4-Gateway username
  - b. Start time
  - c. Interval and overlap
  - d. Services
2. Prepare for receiving notification responses automatically; be aware of the differences between notification responses received via the Notification Broker compared with notification responses to notification requests sent via the AS4-Gateway; see the section "Notification responses" for details.
3. When the requested start time is reached, stop sending notification requests, and start processing the notification responses received from the Notification Broker (via the AS4-Gateway). Notification responses are received in one of the following ways:
  - a. Fetched via pull requests

- b. Received via a push request from the AS4-Gateway to the Economic Operator-controlled AS4 Gateway. See the document “AS4 Gateway - Response messages via push” for more details<sup>(\*)</sup>

(\*) It is recommended to use the push-mode to receive response messages from the AS4-Gateway.

### Daily operation with the Notification Broker

As the Notification Broker only returns non-empty notification responses to the Economic Operator, then there will be periods during the day where the Economic Operator does not receive any notifications.

To determine whether all the relevant notifications have been received, then each notification response from the Notification Broker contains a sequence number. There will be a number sequence per Economic Operator/service, starting with 1, increasing per notification response sent from the Notification Broker.

The following table gives an example of how a series of responses could look, assuming the Economic Operator started to use the notification broker at 2026-02-13T12:01:00, with a period of 120 seconds and an overlap of 10 seconds, and no notifications were generated by DMS in the period 12:00:50 to 12:04:50.

| fromDate            | toDate              | Collaboration ID | Seq no. | Page | Size |
|---------------------|---------------------|------------------|---------|------|------|
| 2026-02-13T12:04:50 | 2026-02-03T12:07:00 | abc-123-def-456  | 1       | 0    | 500  |
| 2026-02-13T12:08:50 | 2026-02-03T12:11:00 | hij-789-klm-012  | 2       | 0    | 500  |
| 2026-02-13T12:08:50 | 2026-02-03T12:11:00 | hij-789-klm-012  | 3       | 1    | 500  |
| ...                 | ...                 | ...              | ...     | ...  | ...  |

Notice the following:

- The collaboration ID will be the same for all pages in a multi-page response.
- No response is received for the period 12:06:50 to 12:09:00, because no notifications were generated in this time interval.
- The sequence number is increased for every notification response sent by the Notification Broker.

### Query Notification Broker status

Before the Notification Broker is deployed to the production environment, there will be provided a service, that Economic Operators can use to query the status of the notification requests/responses handled by the Notification Broker. The exact details of this service will be provided later, but the plan is that it should return the following information for an Economic Operator:

- Last time a notification request was sent by the Notification Broker, and the total size, current page, and total number of pages from the response.
- A list of the last 10 notification responses sent to the Economic Operator.

An Economic Operator can query this service if they are expecting notifications which have not arrived and want to check if the Notification Broker has sent the notification requests for the expected periods and if the responses contained any notifications.

### Behaviour in case of downtime

In case of downtime of the Notification Broker or DMS, then the Notification Broker will start sending the missing notification requests as soon as the systems are operational again. The missing notification requests will be sent continuously after each other with the configured time interval until the toDate catches up with the current time.

In case of downtime of the AS4-Gateway, then the Notification Broker will make the notification responses available for the AS4-Gateway, and once the AS4-Gateway is up and running all the notification responses will be available for the Economic Operator.

## Sending notification requests when using the Notification Broker

Even though the Notification Broker is sending requests on behalf of an Economic Operator, then it is still possible for an Economic Operator to send specific notification requests from their own system. These Economic Operator-generated notification requests will not interfere with the notification requests sent from the Notification Broker.

## Appendix

### Example of an UserMessage element

Below is an example of a UserMessage element in a notification response sent from the AS4-Gateway to the Economic Operator.

```
<eb:UserMessage mpc="urn:fdc:dk.skat.mft.DMS/response/CVR_12345678">
  <eb:MessageInfo>
    <eb:Timestamp>2026-01-30T16:41:39.456Z</eb:Timestamp>
    <eb:MessageId>0d699a23-ca91@DMS.Import2.Notification</eb:MessageId>
  </eb:MessageInfo>
  <eb:PartyInfo>
    <eb:From>
      <eb:PartyId type="string">DMS.Import2.Notification</eb:PartyId>
      <eb:Role>http://docs.oasis-open.org/ebxml-
msg/ebms/v3.0/ns/core/200704/initiator</eb:Role>
    </eb:From>
    <eb:To>
      <eb:PartyId type="string">CVR_12345678_UI_12345678_AS4</eb:PartyId>
      <eb:Role>http://docs.oasis-open.org/ebxml-
msg/ebms/v3.0/ns/core/200704/responder</eb:Role>
    </eb:To>
  </eb:PartyInfo>
  <eb:CollaborationInfo>
    <eb:Service type="string">http://docs.oasis-open.org/ebxml-
msg/as4/200902/service</eb:Service>
    <eb:Action>Response</eb:Action>
    <eb:ConversationId> 00ca27b3-a50a-4d81-9b6d-f1e308c7ead6
  </eb:ConversationId>
  </eb:CollaborationInfo>
  <eb:MessageProperties>
    <eb:Property name="fromDate">2026-02-03T12:13:00</eb:Property>
    <eb:Property name="toDate">2026-02-03T12:16:00</eb:Property>
    <eb:Property name="page">0</eb:Property>
    <eb:Property name="size">500</eb:Property>
    <eb:Property name="sequenceNumber">5</eb:Property>
  </eb:MessageProperties>
  <eb:PayloadInfo>
    <eb:PartInfo href="cid:A1769791299453.34706755@j0AFyd9nKT-CYsIL4kM=">
      <eb:PartProperties/>
    </eb:PartInfo>
  </eb:PayloadInfo>
</eb:UserMessage>
```